

Statement Of Purpose(SOP)

Machine Learning has become one of the most impactful technologies of our time, transforming the way data is analyzed and decisions are made. As a Computer Science Engineering undergraduate with a CGPA of 8.0, I have developed a strong interest in Machine Learning, Data Science, and Data Analytics through my academic coursework, projects, and continuous self-learning. I am excited about the opportunity to participate in the Amazon ML Summer School to strengthen my understanding of machine learning fundamentals and learn from experts working on real-world applications at scale.

My academic curriculum has introduced me to subjects such as Artificial Intelligence and Machine Learning, Data Science, Data Analytics, and Data Structures. These courses have helped me build a foundation in analytical thinking, algorithm design, and data-driven problem solving. Through laboratory work and coursework, I gained exposure to concepts including data preprocessing, feature engineering, model evaluation, and the practical applications of machine learning.

To complement my academic learning, I have worked on several projects that allowed me to apply theoretical concepts to real-world problems. One of my projects was a Face Recognition-Based Attendance System that utilized computer vision techniques to automate attendance tracking. I also worked on a Pothole Detection System using RGB and depth data, where I explored how machine learning and computer vision can be applied to improve road safety and infrastructure monitoring. Additionally, I developed a Data Analysis Platform focused on processing and visualizing datasets to derive meaningful insights. These projects strengthened my skills in problem-solving, data handling, and software development while increasing my interest in intelligent systems.

Beyond academics, I have actively participated in multiple hackathons, where I collaborated with teams to build solutions under time constraints. These experiences improved my ability to learn quickly, work effectively in teams, and translate ideas into functional prototypes. They also exposed me to diverse problem statements and innovative approaches to technology-driven solutions.

I am currently interning at HCLTech, where I am gaining exposure to enterprise technologies and cybersecurity concepts such as Endpoint Detection and Response (EDR), File Integrity Monitoring (FIM), Privileged Access Management (PAM), and related domains. This experience has helped me understand how large-scale systems generate, process, and secure data, further motivating me to explore the role of machine learning in solving complex industry challenges.

I am particularly interested in Amazon ML Summer School because of its focus on building strong machine learning foundations and exposing students to practical applications used in industry. The opportunity to learn from researchers and engineers at Amazon would provide

valuable insights into state-of-the-art machine learning techniques and their deployment at scale.

My long-term goal is to build expertise in machine learning and data-driven technologies and contribute to developing intelligent solutions that create meaningful impact. I believe Amazon ML Summer School will be an important step in this journey by enhancing my technical knowledge, strengthening my problem-solving abilities, and preparing me for future opportunities in machine learning research and development.